

Full Stack Web Development Lab

Week 3 – Record

Aim

To design a "Student Name Manager" web application using HTML, CSS and JavaScript that allows a user to add student names to a list, view the total number of students, and delete any student from the list, using JavaScript DOM manipulation.

Description

This lab demonstrates dynamic web pages using JavaScript along with HTML and CSS. An array (**students**) is used to store the list of student names in memory. The **addStudent()** function reads the value typed in the input box, validates that it is not empty, capitalizes the first letter, and pushes it into the array. The **displayStudents()** function then rebuilds the `<ul>` list on the page using **innerHTML**, showing each student's serial number, name, and a Delete button. The **deleteStudent(index)** function removes a student from the array using **splice()** and refreshes the list. The page also shows a live count of "Total Students" that updates automatically. CSS is used separately to style the card layout, input box, list items, and buttons.

Program

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Student Name Manager</title>
  <link rel="stylesheet" href="login.css">
</head>
<body onload="displayStudents()">

  <div class="container">
    <h1>Student Name Manager</h1>

    <div class="input-box">
      <input type="text" id="studentName" placeholder="Enter student name">
      <button onclick="addStudent()">Add Student</button>
    </div>

    <h2>Total Students: <span id="count">0</span></h2>

    <ul id="studentList"></ul>
  </div>

  <script src="login.js"></script>
</body>
</html>
```

login.css

```
*{
  margin:0;
  padding:0;
  box-sizing:border-box;
  font-family:Arial, Helvetica, sans-serif;
}

body{
  background:#f4f4f4;
  display:flex;
```

```

        justify-content:center;
        align-items:flex-start;
        min-height:100vh;
        padding:50px 20px;
    }

    .container{
        width:600px;
        background:#fff;
        padding:30px;
        border-radius:12px;
        box-shadow:0 5px 20px rgba(0,0,0,.15);
    }

    h1{
        font-size:42px;
        margin-bottom:25px;
    }

    .input-box{
        display:flex;
        gap:10px;
        margin-bottom:25px;
    }

    .input-box input{
        flex:1;
        padding:12px;
        font-size:18px;
        border:1px solid #bbb;
        border-radius:3px;
        outline:none;
    }

    .input-box input:focus{
        border-color:#4a90e2;
    }

    .input-box button{
        padding:12px 18px;
        font-size:17px;
        cursor:pointer;
        border:1px solid #999;
        background:#f8f8f8;
        border-radius:3px;
        transition:.3s;
    }

    .input-box button:hover{
        background:#e8e8e8;
    }

    h2{
        margin-bottom:20px;
        font-size:30px;
    }

    #studentList{
        list-style:none;
    }

    #studentList li{
        background:#eef0ff;
        margin-bottom:12px;
        padding:15px;
        border-radius:5px;
        display:flex;
        justify-content:space-between;
        align-items:center;
        font-size:22px;
    }

    .delete{
        background:#e74c3c;
        color:#fff;
        border:none;
        padding:10px 18px;
    }

```

```

border-radius:4px;
cursor:pointer;
font-size:16px;
}

.delete:hover{
background:#c0392b;
}

```

login.js

```

let students = ["Bob"];

function addStudent() {
  const input = document.getElementById("studentName");
  let name = input.value.trim();

  if (name === "") {
    alert("Enter student name");
    return;
  }

  // Capitalize first letter
  name = name.charAt(0).toUpperCase() + name.slice(1).toLowerCase();

  students.push(name);

  input.value = "";
  displayStudents();
}

function displayStudents() {
  const list = document.getElementById("studentList");
  list.innerHTML = "";

  students.forEach((student, index) => {
    list.innerHTML += `
    <li>
      <span>${index + 1}. ${student}</span>
      <button class="delete" onclick="deleteStudent(${index})">
        Delete
      </button>
    </li>`;
  });

  document.getElementById("count").textContent = students.length;
}

function deleteStudent(index) {
  students.splice(index, 1);
  displayStudents();
}

displayStudents();

```

Details about the Program

Where Saved	The three files are saved together inside the project folder FL_LAB_ANITS/fs_week_3/ in the GitHub repository raviteja-codehub/FL_LAB_ANITS — index.html (structure), login.css (styling), and login.js (logic).
Where Executed	The page runs inside a web browser (Google Chrome / Edge / Firefox). It can also be hosted live using GitHub Pages at: https://raviteja-codehub.github.io/FL_LAB_ANITS/fs_week_3/

How Executed

1. Locally: Open **index.html** by double-clicking it or via "Open with Browser". The browser loads **login.css** for styling and **login.js** for functionality (linked via `<link>` and `<script>` tags). Typing a name and clicking "Add Student" runs **addStudent()**; clicking "Delete" runs **deleteStudent()**.
2. Online: The folder is pushed to GitHub and served live through GitHub Pages, so the same interaction works directly from the browser without downloading any files.

Output

On opening index.html in the browser, entering names, and clicking "Add Student", the list updates live as shown below:

Student Name Manager

Add Student

Total Students: 3

1. BobDelete

2. Raviteja pasupuletiDelete

3. PrasanthiDelete